Online Appendix

Michael Ryan and Ayumu Tanaka "Anchored Abroad: Sunk Costs, Asset Specificity, and Japanese Affiliate Exit from Brexit Britain"

X1 Affiliate size of exiting affiliates

Figure X1 presents the number of exiting affiliates by size category. While exits are most concentrated among the smallest-size affiliates, a considerable number of exits are also observed in the medium- and large-size categories.

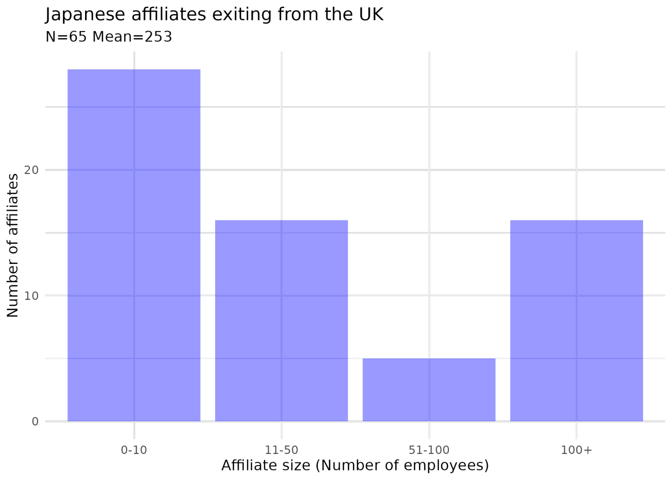

Figure X1: Affiliate size of exiting affiliates

Source: Authors' compilation based on Toyo Keizai's OJC data.
Note: The figure presents the number of affiliates by size category, excluding those without employee data. In contrast, the regression analysis in the main text includes these affiliates by imputing missing employee numbers.

X2 Average ownership ratio

Figure X2 presents the average ownership ratio by country group. The UK exhibits a higher average ownership ratio than high-income EU member countries. The figure also indicates an upward trend in ownership ratios in both groups. However, it does not provide clear evidence that the gap between the UK and high-income EU member countries is widening.

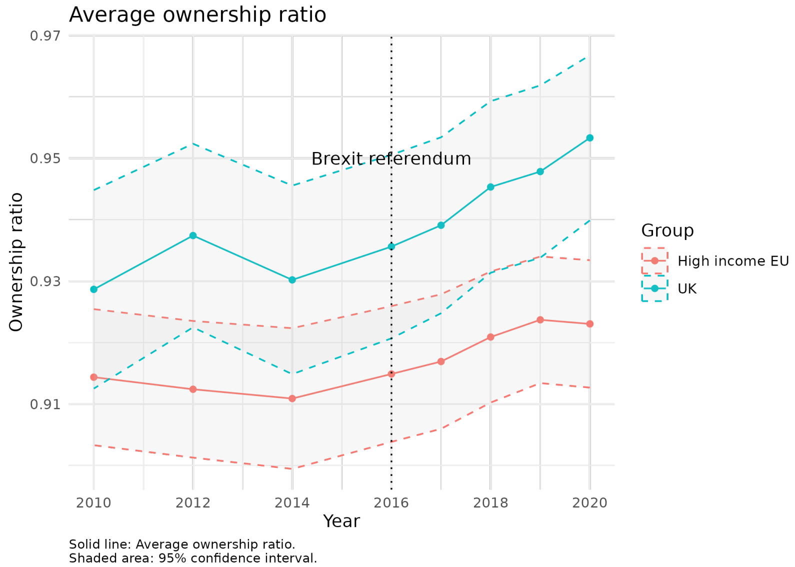

Figure X2: Average ownership ratio by country group

Source: Authors' compilation based on Toyo Keizai's OJC data.
Note: The figure presents the evolution of average ownership by country group, along with 95% confidence intervals. It compares the average ownership ratio in high-income EU member countries with that in the UK.

X3 Number of firms and affiliates exiting the UK by sector

Figure X3 shows the number of firms and affiliates exiting the UK by sector. It indicates that firm exits are relatively concentrated in the manufacturing sector, whereas affiliate exits are more prevalent in the services sector.

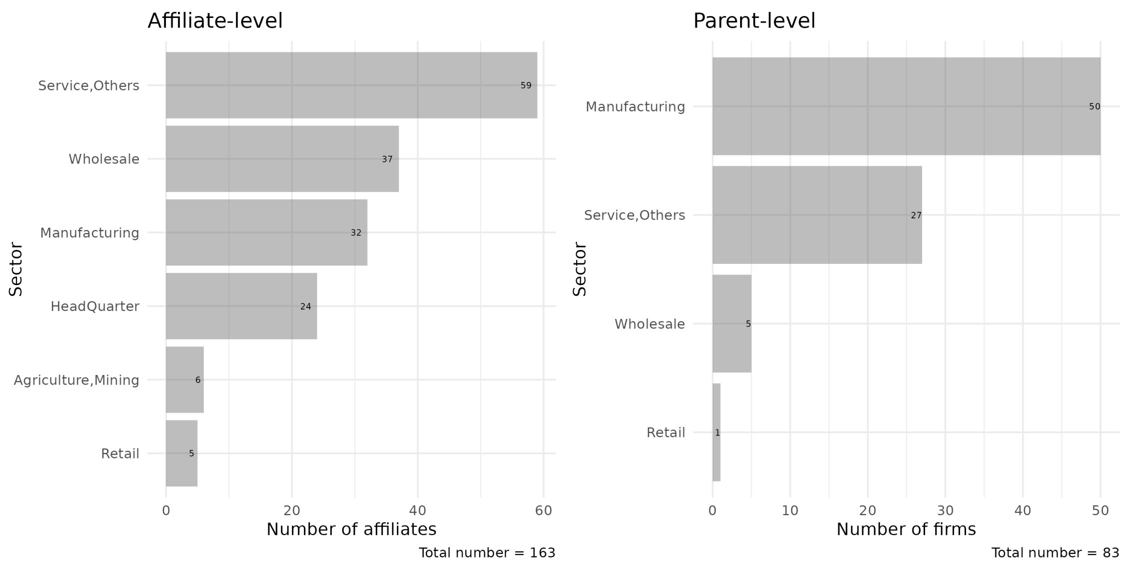

Figure X3: Number of firms and affiliates exiting the UK by sector: 2016–2020

X4 Kaplan–Meier survival plots

While Figure 2 of the main text suggests no significant impact of Brexit on exit, we wish to more robustly examine this data. To begin, we employ Kaplan-Meier estimation to study Japanese exits from the UK. This method enables us to assess the survival rates of different affiliate cohorts categorized by distinct characteristics. In this case, affiliate location is the distinct characteristic, as we utilize affiliate location data from the OJC to compare the survival rates of Japanese affiliates in the UK with those in the remaining EU member countries. Figure X4 illustrates the Kaplan-Meier estimates for both groups from 1990 to 2021. In the figure, year 0 represents affiliate birth, with firms still in existence in 2021 having (right-censored) ages at 31 years. Both curves exhibit the expected downward path dependence, indicating declining affiliate survival likelihood (or increased exit likelihood of exit) as affiliates age. Note that the curve representing UK-based affiliates lies below that for the EU-based affiliates. While this suggests a higher probability of exit, the figure suggests that there is not much difference between the two groups. In fact, results from a Mantel-Cox log-rank test indicate a p-value of 0.1, confirming no significant survival difference for Japanese affiliates in our two groups.

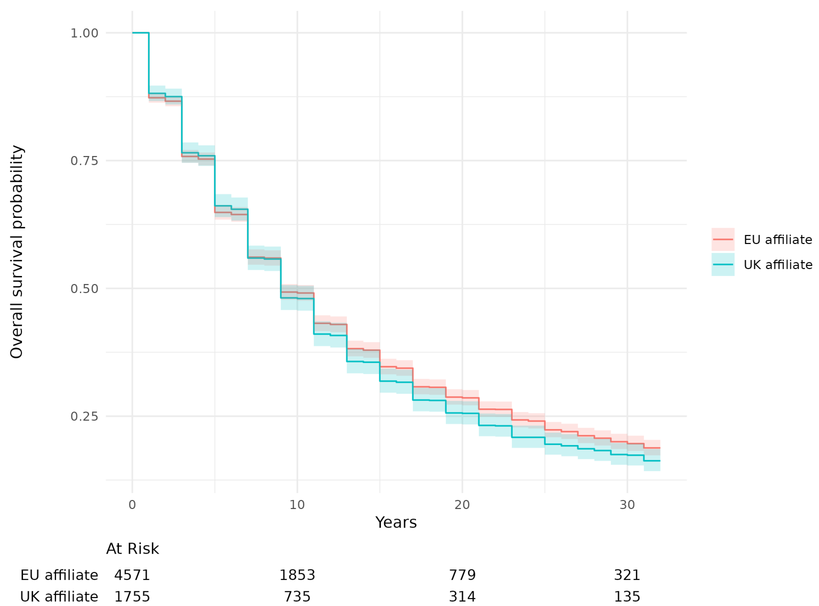

Figure X4: Kaplan–Meier survival estimates: UK versus EU member countries

Source: Authors' compilation based on Toyo Keizai's OJC data.
Note: This figure compares the survival curve of Japanese affiliates in the UK and that in other EU member countries.

Table X1: Cox regression model at affiliate-level

	World		Europe only	
	(1)	(2)	(3)	(4)
	w/o country FEs	with country FEs	w/o country FEs	with country FEs
Brexit	1.049 26	1.097 83	1.048 18	1.042 27
	(0.083 17)	(0.087 09)	(0.087 53)	(0.087 22)
Ownership ratio	1.228 95***	1.194 91***	0.827 30**	0.819 18**
	(0.026 49)	(0.025 10)	(0.051 41)	(0.051 05)
Affiliate size	0.999 95***	0.999 95***	0.999 95	0.999 95
	(0.000 01)	(0.000 01)	(0.000 04)	(0.000 04)
GBR	0.934 95*		0.947 71	
	(0.027 17)		(0.032 08)	
log GDP, PPP	1.025 77***		0.937 31***	
	(0.003 78)		(0.015 53)	
log GDP per capita, PPP	0.892 71***		0.632 00***	
	(0.006 27)		(0.047 20)	
log Distance from Japan	0.896 44***		0.353 47***	
	(0.007 70)		(0.106 58)	
Fixed Effect: Year	YES	YES	YES	YES
Fixed Effect: Sector	YES	YES	YES	YES
Fixed Effect: Country	NO	YES	NO	YES
Num.Obs.	43 597	45 826	6174	6174

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note:
Hazard ratios are reported. The Cox proportional hazards model estimates them. Failure event is exit. Variables with hazard ratios greater than 1 are associated with a higher likelihood of exit and shorter survival time. Variables with a hazard ratio smaller than 1 are associated with a lower likelihood of exit and longer survival time. Host countries' fixed effects and year-fixed effects are included in the estimation. Standard errors are presented in blankets.

X5 Cox proportional hazards model

We use the Cox proportional hazards model (Cox, 1972) to explore the factors affecting affiliate exits.

The hazard at time t is assumed to be:

$$h(t) = h_0(t) \exp\left(\beta_1 Brexit + \beta_2 x_2 + \cdots + \beta_k x_k\right) \tag{1}$$

where h_0 is the baseline hazard and $\beta_1, ..., \beta_k$ are covariates. The Cox model provides estimates of coefficients $\beta_1, ..., \beta_k$. Our key variable, *Brexit*, takes a value of one if the host country is the UK and $t \geq 2016$. A failure event is an exit in our analysis. Variables with hazard ratios greater than 1 are associated with a higher likelihood of exit and shorter survival time. Variables with a hazard ratio smaller than 1 are associated with a lower likelihood of exit and longer survival time. The estimation results in Table X1 show that Brexit did not have a significant impact on the Japanese affiliates' survival in the UK.

Figure X5 presents coefficient estimates and 95% confidence intervals of the Cox proportional hazard model. Models 1–4 of Figure X5 correspond to estimation results in columns (1)–(4) of Table X1, respectively. Figure X5 reveals that the 95% confidence interval of the Brexit dummy is quite large across all models, suggesting that the impact of Brexit is insignificant and heterogeneous across Japanese affiliates in the UK. The Cox model cannot include either affiliate fixed effects or sector $\times$ year fixed effects because they lead to the failure of convergence. For this reason, in the main text, we use the linear probability model that includes affiliate fixed effects and year fixed effects. The linear probability model results are more reliable because they control for affiliate heterogeneity and time heterogeneity. However, both models indicate that the impact of Brexit is trivial.

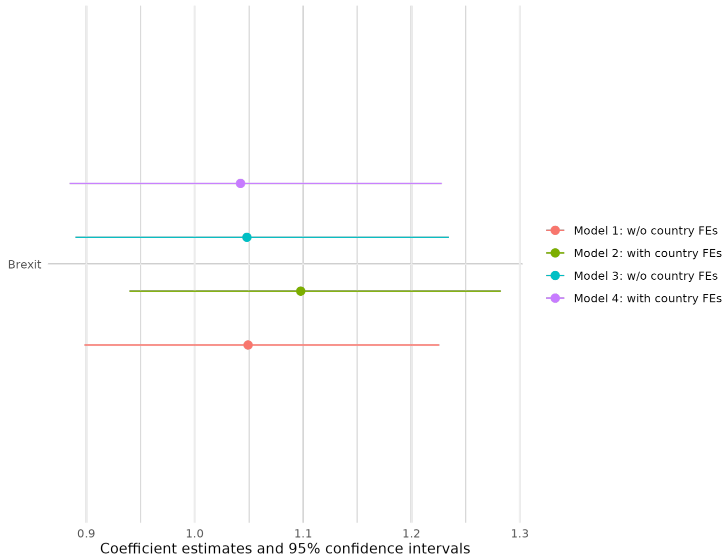

Figure X5: The Cox regression model's estimates of the Hazard ratio

References

Cox, David R, "Regression models and life-tables," *Journal of the Royal Statistical Society: Series B (Methodological)*, 1972, *34* (2), 187–202.